

LOADCAST

Thirty days of wearable data, read as a physiological trajectory — not a summary.

Predicting **whether** an athlete breaks down in the next month, **when** it starts, and **how long** they are sidelined — built strictly from the observation window and validated against the competition's own scoring function.

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INJURY IS A PROCESS, NOT AN EVENT

Most tabular approaches summarise a month into averages. But an athlete who trained steadily and one who ramped hard in the final week can have **identical means** and completely different risk. Our thesis: the **shape** of the 30 days carries the signal, and the shape is measurable.

HYPOTHESIS	TESTABLE	PREDICTION IF TRUE	FALSIFIABLE	VERDICT	CONFIRMED
LOAD OUTFRUNS ADAPTATION		A VISIBLE PRE-RAMP		THE DATA AGREES	
Overuse injury follows a mismatch between workload accumulation and the body's capacity to adapt. That mismatch leaves fingerprints: rising acute-to-chronic load, falling training variation, autonomic drift, and shrinking recovery.		If the thesis holds, injured athletes should show a measurable upward load trajectory in the observation window that healthy athletes do not — and the <i>steepness</i> of that ramp should encode how soon the breakdown arrives.		Aligned on true onset, mean load climbs 324 → 411 then falls off a cliff. Observation-window features reach r = 0.855 against onset day. Slide 04 shows it.	

WHAT MAKES THIS APPROACH DIFFERENT	IDEATION	WHY IT MATTERS BEYOND THE LEADERBOARD
1 Scale-free, not population-normalised. Every core feature is a ratio or a slope measured against the athlete's <i>own</i> baseline. One model serves a goalkeeper and a sprinter without per-sport retraining. 2 The scoring function is the spec. We implemented the brief's metric — penalty, baseline, skill — as a validation function <i>before</i> writing a model, then argued every decision against that number. 3 Physiology, not just statistics. Each feature family encodes a published injury mechanism, so importances are readable by a coach and the output names a cause, not just a risk.		SCOPE OF ROUND 1 Ideation plus a working end-to-end prototype: 100 features, three model heads, cross-validated against the official metric, and two valid submissions generated. Round 2 turns the idea into a survival model and a staff-facing board.

THREE PREDICTIONS, ONE HIDDEN MONTH

02

Days 1–30 are visible; days 31–60 are not. The two timing numbers are graded on **every athlete who is truly injured** — including the ones we said were fine.

TARGET A	F1	TARGET B	MAE → SKILL	TARGET C	MAE → SKILL
INJURED_IN_RISK_WINDOW		ONSET_DAY_OFFSET		RECOVERY_DURATION	
Binary. Does an onset fall inside days 31–60? Both false alarms and misses cost.		Integer 1–30. Which day the injury begins, scored against a mean-prediction baseline.		Days sidelined after onset. Required for every athlete — even those predicted healthy.	

THE FOUR TRAPS	WHAT ACTUALLY MAKES THIS HARD	METRIC, IMPLEMENTED LITERALLY	VALIDATION HARNESS
1 Asymmetric penalty. A missed injury books 30 days of error on <i>both</i> timing heads — 3.9× the onset baseline MAE. One miss cannot be repaid by a hundred good timing predictions. 2 Train ships 60 days; test ships 30. The extra month is the answer key, and a leakage trap that would inflate validation and collapse on the leaderboard. 3 Conditional targets, unconditional scoring. Regressors can only learn from injured athletes, yet must fire for everyone. 4 Answers must cohere. A confident risk call with an incoherent onset day is not a usable recommendation.		■ Hit (predicted 1, truly 1): MAE against true onset and recovery. ■ Miss (predicted 0, truly 1): fixed PENALTY = 30 on both. ■ Skill = <code>max(0, 1 - MAE_model / MAE_baseline)</code>. ■ Baseline = training-set mean onset (7.61 MAE) and mean recovery (3.24 MAE). CONSEQUENCE WE DESIGNED AROUND The two heads die at different speeds: recovery skill is gone once recall falls below ~99%, onset skill once it falls below ~81%. Nothing short of near-total recall keeps both alive — which is why our optimum ends up degenerate. Slide 08.	

PROFILING TEN TABLES AND THREE TARGETS

03

4,100 athletes, 19.4 M hourly readings, 213 K athlete-days. Before modelling we profiled coverage, distributions and target shape — which is where the first strategic finding appeared: **the obvious risk factors carry almost no signal.**

3,000

TRAIN ATHLETES

1,100

TEST ATHLETES

35.0%

INJURY BASE RATE

15.3

MEAN ONSET DAY

11.5

MEAN RECOVERY DAYS



FINDINGS THAT SHAPED THE MODEL

EDA

- **Static risk factors are nearly useless.** Injury rate varies only 0.322–0.365 across six sports, and prior-season injuries run the *wrong* way (0→.355, 1→.344, 2→.322) before a noisy jump at 3 (.492, n = 63). The signal is not demographic.
- **Onset is near-uniform over 1–30**, so the mean-prediction baseline is weak (MAE 7.61) and genuinely beatable.
- **Recovery is bell-shaped but unconditioned** — its strongest correlate anywhere in 100 features reaches only $|r| = 0.058$.
- **Class balance is workable** at 35% positive; no resampling needed.

WHERE WE LOOKED NEXT

If who the athlete *is* does not predict injury, then what the athlete *did* must. That pushed the entire feature effort into temporal dynamics.

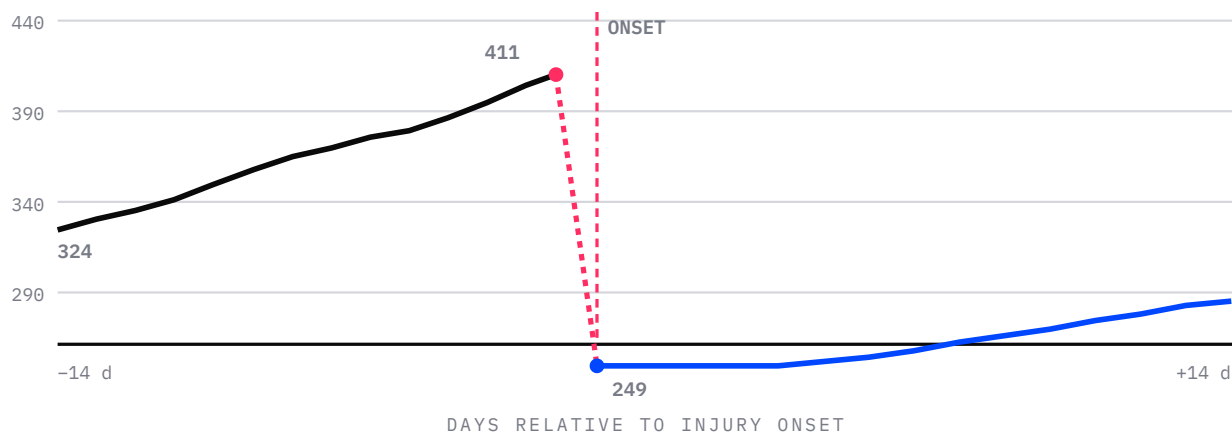
INJURY ANNOUNCES ITSELF IN ADVANCE

04

Averaging daily load across all 1,050 injured athletes, aligned on each athlete's true onset day, exposes the mechanism: a **ramp running at least two weeks**, then a cliff. This is the pattern the features are built to catch while it is still inside the observation window.

MEAN DAILY LOAD, ALIGNED ON ONSET

N = 1,050



— PRE-INJURY RAMP — ONSET CLIFF — RECOVERY CLIMB

SEPARATION INSIDE THE OBSERVATION WINDOW

DAYS 1-30 ONLY

- **Acute:chronic ratio 1.051 vs 1.000.** Injured athletes end the month *accelerating*; healthy ones are flat. Small in absolute terms, highly consistent across 3,000 athletes.
- **Onset correlates up to $|r| = 0.855$** with single observation-window features. The ramp's steepness encodes how far away the breakdown is.
- **Recovery correlates with nothing** — best feature $|r| = 0.058$. We report this rather than pretend otherwise.

THE COUNTER-INTUITIVE RESULT

When the injury lands is substantially easier to predict than *whether* it lands — the inverse of the usual assumption, and the reason our scoring strategy protects timing above all else.

CLIP, CLEAN, COLLAPSE, PRUNE

05

Ten tables at four different grains become **one row per athlete**. The order matters: clip the window first, then repair quality issues, then aggregate, then delete anything that cannot carry information.

1 • CLIP

Every source hard-filtered to 2026-01-05 → 02-03



2 • PARSE

Three datetime formats normalised to one



3 • COLLAPSE

Hourly → daily → athlete, 19.4 M → 4,100 rows



4 • IMPUTE

Sparse sources left as NaN for LightGBM



5 • PRUNE

8 dead columns dropped: 108 → 100

QUALITY AUDIT

WHAT WE FOUND

- **SedentaryMinutes is constant at 1200** for all 213 K rows — zero variance, and it silently produced four dead features.
- **TrackerDistance equals TotalDistance** in 100% of rows — perfectly redundant.
- **TotalSleepRecords is always 1.**
- **No nulls and no duplicate athlete-days** in the daily or sleep tables.

MISSINGNESS

HANDLED, NOT DROPPED

- **weightLogInfo covers 83.5% of athletes**, median 3 logs each; the `Fat` column is **69.4% null**.
- We keep the source and let LightGBM route NaN natively — discarding it would drop a sixth of the cohort's body-composition signal.
- The RandomForest head, which cannot take NaN, gets a median imputer inside its pipeline.
- Every other source has **100% athlete coverage**.

PRUNING

108 → 100 FEATURES

Zero-variance and exactly-duplicated columns are detected **on the training set only** and removed from both sets.

- 7 constant: three `SedentaryMinutes_*` columns plus `sed_slope`, `rest_days`, `hours_max`, `sessions_per_active_day`
- 1 duplicate: `active_days`

MEASURED, ONE VARIABLE AT A TIME

Re-running the *identical* pipeline with `--no-prune` isolates it: AUC **+0.005**, F1 **+0.010**, but recovery skill **-0.006** and composite **-0.002**. We keep it because F1 is an officially scored target and our composite is not.

100 FEATURES FROM SPORTS SCIENCE

Each family encodes a published mechanism of overuse injury — **load accumulation, insufficient variation, autonomic drift, recovery debt, schedule congestion** — which is why the resulting importances are interpretable to a coach rather than a black box.

FAMILY 01LOAD & ACWR Internal load proxy <code>3×very + 2×fairly + 1×light</code> active minutes, then acute:chronic ratios at 7:28 and 3:21. ■ Weekly means, W4/W1 ratio, largest week-over-week jump ■ Spike days above 1.5× personal median	FAMILY 02MONOTONY & STRAIN Mean load ÷ its standard deviation. Grinding the same session daily with no hard/easy variation is a precursor <i>independent of volume</i>. ■ Strain = total load × monotony ■ Personal-baseline z-score of the peak day	FAMILY 03AUTONOMIC Nocturnal resting HR reconstructed from the 01:00–05:00 band, then its slope across the month. Rising RHR under constant load = non-adaptation. ■ Last-7 vs whole-window delta ■ HR reserve trend, daily max drift
FAMILY 04RECOVERY Cumulative sleep shortfall against an 8-hour reference, efficiency (asleep ÷ in bed), nights under six hours, final-week trend against the month.	FAMILY 05CONGESTION Longest streak of consecutive training days without rest, sessions per active day, scrimmage-vs-gym mix, dispersion of session start times as a circadian proxy.	FAMILY 06CONTEXT & CROSSES Sport, position, team, BMI, experience-per-age, prior injuries per season — crossed with load to give <code>load_per_sleep</code>, <code>acwr×prior_inj</code>, <code>rhr_rise×load</code>.

SIX MECHANISMS, ONE LATENT – THE HONEST READING

We engineered six independent mechanisms; **this dataset contains roughly one**. Monotony, ACWR, RHR drift and load slope inter-correlate at $|r| = 0.80\text{--}0.99$, median **0.92** (`monotony` vs `strain`: **0.9995**), 70 pairs table-wide exceed 0.98, and one principal component holds **92.8%** of that block — which is why four unrelated features all land at $|r| \approx 0.855$ against onset. Onset MAE by how much of it you use: **7.61** baseline → **3.43** one feature → **2.92** PC1 → **2.64** all 95 (2.61 with the shipped blend).

CHOSEN ON EVIDENCE, NOT ON HABIT

Every candidate ran on the **same feature table and the same metric**. Gradient boosting won, but narrowly — and that narrowness is itself the finding.

TASK A — CLASSIFICATION		5-FOLD OOF
MODEL	ROC-AUC	BEST F1
Decision tree (d=6)	0.735	0.631
LightGBM + native cats	0.752	0.650
Random forest	0.760	0.651
Logistic regression	0.761	0.648
LightGBM, numeric only	0.762	0.654
Final config CHOSEN	0.767	0.657

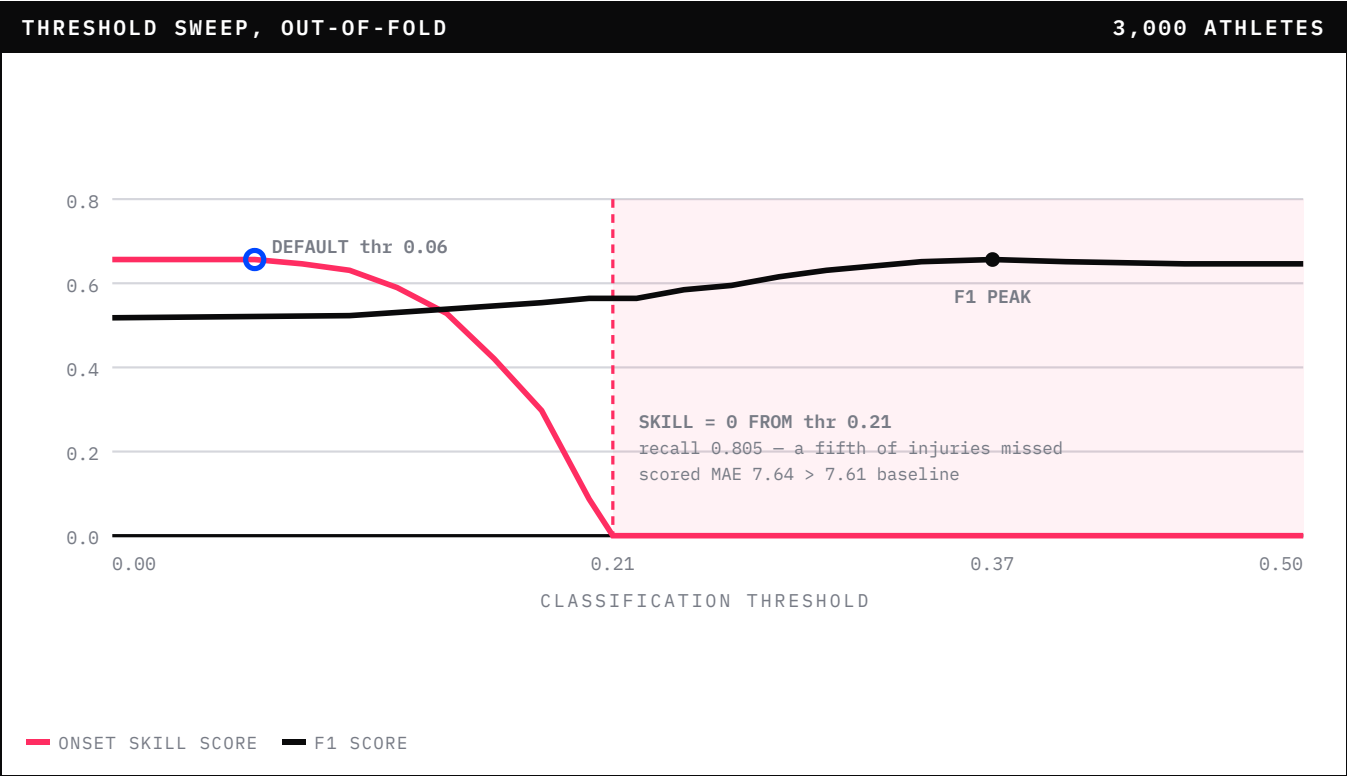
TASK B — ONSET REGRESSION		INJURED ONLY
MODEL	MAE	SKILL
Mean baseline (brief)	7.615	0.000
Ridge regression	2.971	0.610
LightGBM (L2 loss)	2.760	0.638
LightGBM (L1 loss)	2.658	0.651
Random forest	2.623	0.656
LGBM(L1) + RF blend CHOSEN	2.610	0.657

FINAL ARCHITECTURE	THREE HEADS, ONE TABLE
■ Head A — LightGBM classifier, 31 leaves, lr 0.03, L2 = 5, early stopping on fold AUC. All 3,000 athletes; categoricals passed natively, categories fixed across train and test. ■ Heads B & C — $0.6 \times \text{LightGBM(L1)} + 0.4 \times \text{RandomForest}$, fit on the 1,050 injured only. L1 objective deliberately: the metric is MAE, so the loss <i>is</i> the metric. ■ Ensembling — averaged across the five folds; the saved artefact <i>is</i> that ensemble, so inference reproduces the validated numbers exactly.	

THE NARROWNESS IS THE FINDING Logistic regression reaches 0.761 vs LightGBM's 0.767. Not “great features” — slide 06: the predictors are near-collinear, so there is one signal and every model finds it.
READ THE COMPARISON HONESTLY Benchmark rows use the unpruned 108-feature table at fixed 600 trees; the final row adds pruning and early stopping. Native cats <i>lose</i> 0.010 AUC at fixed depth — early stopping recovers it.

THE PENALTY INVERTS THE USUAL TRADE-OFF

Chasing the best F1 is the obvious move, and it is the wrong one. Because a missed injury costs 30 days on both timing heads, **recall is not a preference here — it is a budget constraint**, and we can compute exactly where it binds.



THE ARITHMETICWHY 0.21 IS A CLIFF

Measured, not illustrated: at threshold **0.21** recall is **0.805**, and the scored onset MAE is **7.643** against a 7.615 baseline. **Skill is already exactly zero there** — a fifth of the injuries missed is enough to erase every correct timing prediction.

Moving from the F1 peak at 0.37 down to 0.06 costs **0.138 F1** and buys back **0.657** onset skill plus **0.055** recovery skill. A five-to-one return in the units the leaderboard adds up.

DECISIONTWO SUBMISSIONS, ONE DEFAULT

OPERATING POINT	RECALL	F1	COMP.
Recall-first DEFAULT	1.000	0.519	0.411
F1-balanced ALT	0.520	0.657	0.219

Full disclosure: at threshold 0.06 the optimum is degenerate — it flags *every* athlete, so the classifier contributes nothing to the score. We ship it because it maximises the metric as written, and we ship the F1-balanced file alongside in case Task A is graded on its own.

OUT-OF-FOLD, BOTH WAYS SCORED

Every number is 5-fold out-of-fold on 3,000 athletes. Timing is reported **twice**: on hits only, and with the 30-day miss penalty applied — because those differ everywhere except at our operating point, and hiding the difference would undercut the whole argument.

0.767

ROC-AUC

0.754

PR-AUC

0.657

BEST F1 @ THR 0.37

2.61

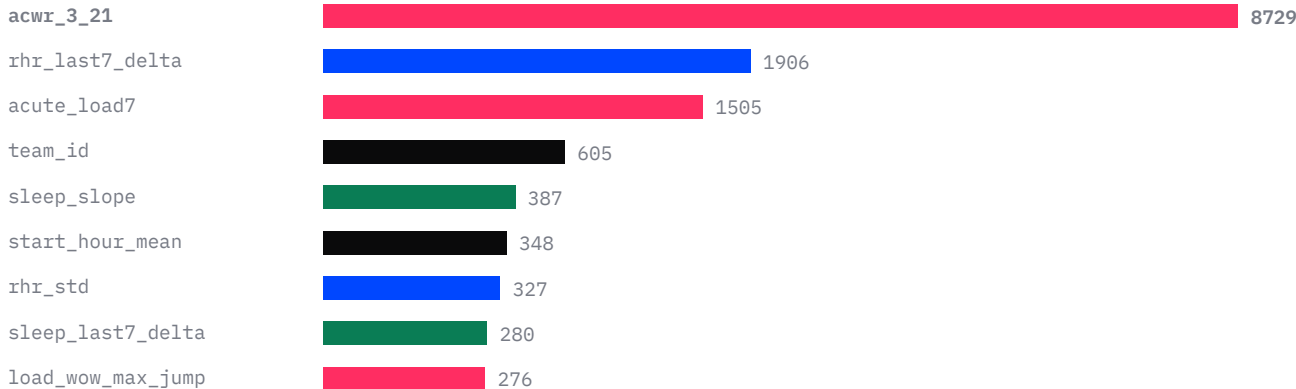
ONSET MAE, DAYS

0.411

COMPOSITE @ DEFAULT

TOP FEATURES BY GAIN

BAR LENGTH ∝ √GAIN



LOAD RATIO AUTONOMIC RECOVERY CONTEXT

SKILL: HITS-ONLY VS PENALTY-INCLUDED

INJURED ATHLETES ONLY

TARGET	BASE MAE	HITS MAE	HITS SKILL	SCORED @0.06	SCORED @0.37
onset	7.61	2.61	0.657	0.657	0.000
recovery	3.24	3.06	0.055	0.055	0.000

Equal at 0.06 *only because* recall is 1.000. At the F1 peak, half the injuries are missed and **both skills collapse to zero**.

WHAT WE ARE NOT CLAIMING

HONEST LIMITS

- **Recovery is near-unlearnable** — best $|r| = 0.058$, so 0.055 is near the ceiling.
- **The default flags every athlete** — the composite optimum is degenerate and the classifier earns nothing at it.
- **Pruning is not free**: -0.002 composite for +0.010 F1, an official target.
- **No test labels exist** — all figures are cross-validated.

FROM LEADERBOARD TO TOUCHLINE

A ranked risk board is more useful to a coach than a binary label. The same model that produces the submission produces a **daily watchlist with the reason attached** — and every feature it uses is one a coach can act on.

PRACTICALITY	RUNS TODAY	SCALABILITY	LINEAR	IMPACT	DECISIONS IT CHANGES
DEPLOYABLE NOW ■ End-to-end in ~6 min on a laptop CPU; no GPU, no external services ■ pandas + LightGBM + scikit-learn only ■ Inference for a new athlete is a single 100-column row		SCALES BY ATHLETE ■ Feature build is O(n) and per-athlete independent — trivially parallel across a squad or a league ■ Rolling windows recompute incrementally; a nightly job replaces the batch ■ Saved ensemble is a few MB — deployable on-prem at a club		WHO ACTS ON IT ■ Rank the squad by risk, then deload the top decile before the ramp completes ■ Onset to ±2.6 days lets medical staff pre-book scans and plan rotation ■ Attribution names the cause: ramp, monotony, RHR drift or sleep debt	

ROUND 2 BUILD PLAN		PRIORITISED
1	Discrete-time hazard model. Onset day is a time-to-event target. A per-day hazard over days 31–60 makes Tasks A and B <i>the same model</i> — the survival curve yields both the probability and the expected day, so consistency is guaranteed rather than patched on.	
2	Quantile regression for recovery. Given the weak signal, an 80% interval beats a point estimate.	
3	Model the latent directly. Correlation pruning is tested and <i>rejected</i> — it costs up to 0.011 AUC and never improves onset MAE; best case, unchanged. The open lever is the reverse: estimate the shared ramp factor explicitly, plus a sequence encoder over the raw 30×24 grid.	

ALSO QUEUED	
Per-sport calibration · SHAP · group-aware CV · seed ensembling. Ask the organisers: are Tasks A and B combined or scored separately? That decides the primary file.	

SUBMISSION PACKAGE	REPRODUCIBLE
■ Pipeline — <code>build_features</code>, <code>train_model</code>, <code>predict</code> ■ Evidence — <code>eda_profile</code>, <code>benchmark_models</code>, <code>collinearity</code>, <code>--no-prune</code> ablation ■ Artefacts — saved ensemble, both submissions, metrics JSON, deck as PDF + PPTX via <code>export_deck</code>	